# RAG Documentation Chatbot - Sample Document

## Overview

This sample document describes how the RAG Documentation Chatbot works. RAG stands for Retrieval-Augmented Generation: the app retrieves relevant passages from your uploaded PDF and then generates an answer grounded in those passages.

## How it works

First, an uploaded PDF is split into small text chunks. Each chunk is turned into an embedding, a list of numbers that captures its meaning. When you ask a question, the app finds the chunks closest to your question and sends them to the language model, which writes an answer using only those chunks.

## Main features

Upload a PDF to create a private, per-session knowledge base. Ask questions in plain English. Every answer lists the supporting passages with their page numbers. If the document does not contain the answer, the chatbot says so instead of guessing.

## Technology

The backend is a FastAPI service. The frontend is a React app built with Vite and hosted on Vercel. Embeddings and answers are generated by Google's Gemini API, and chunks are stored in ChromaDB.

## Contact

This project is maintained by Laya Khathoon as a portfolio project.